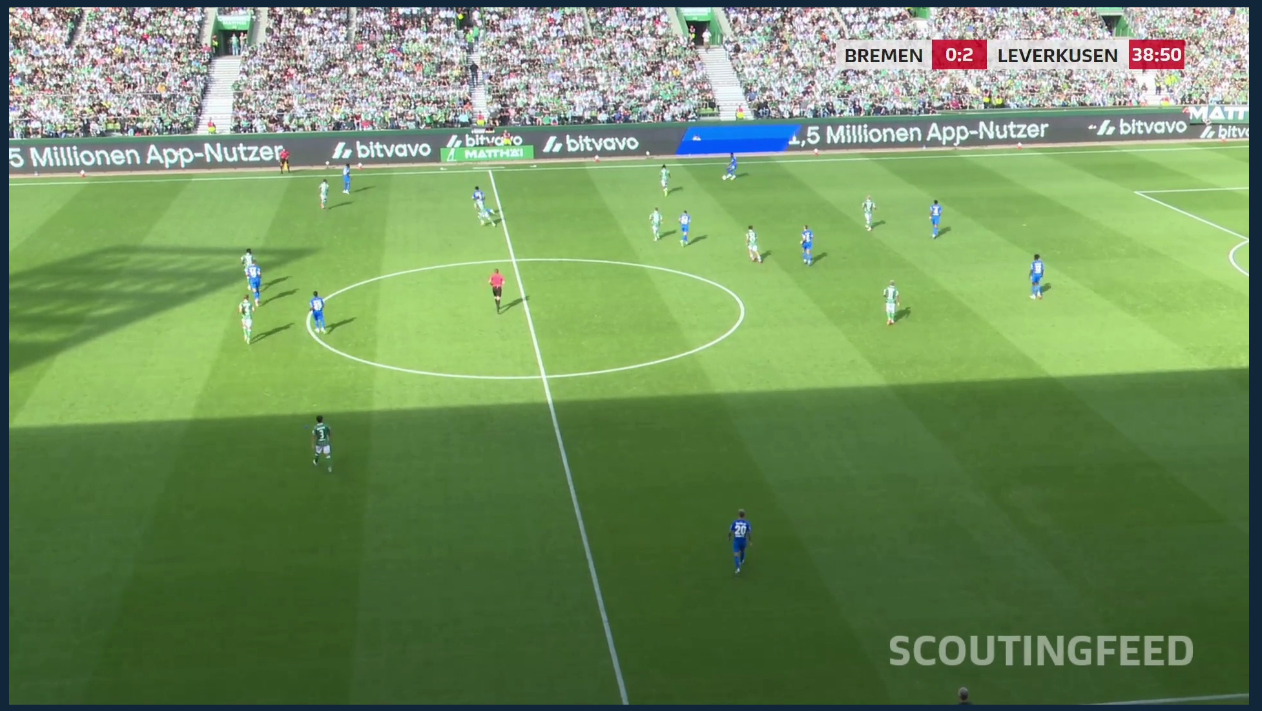
Preprocessing effects on fixed pose models

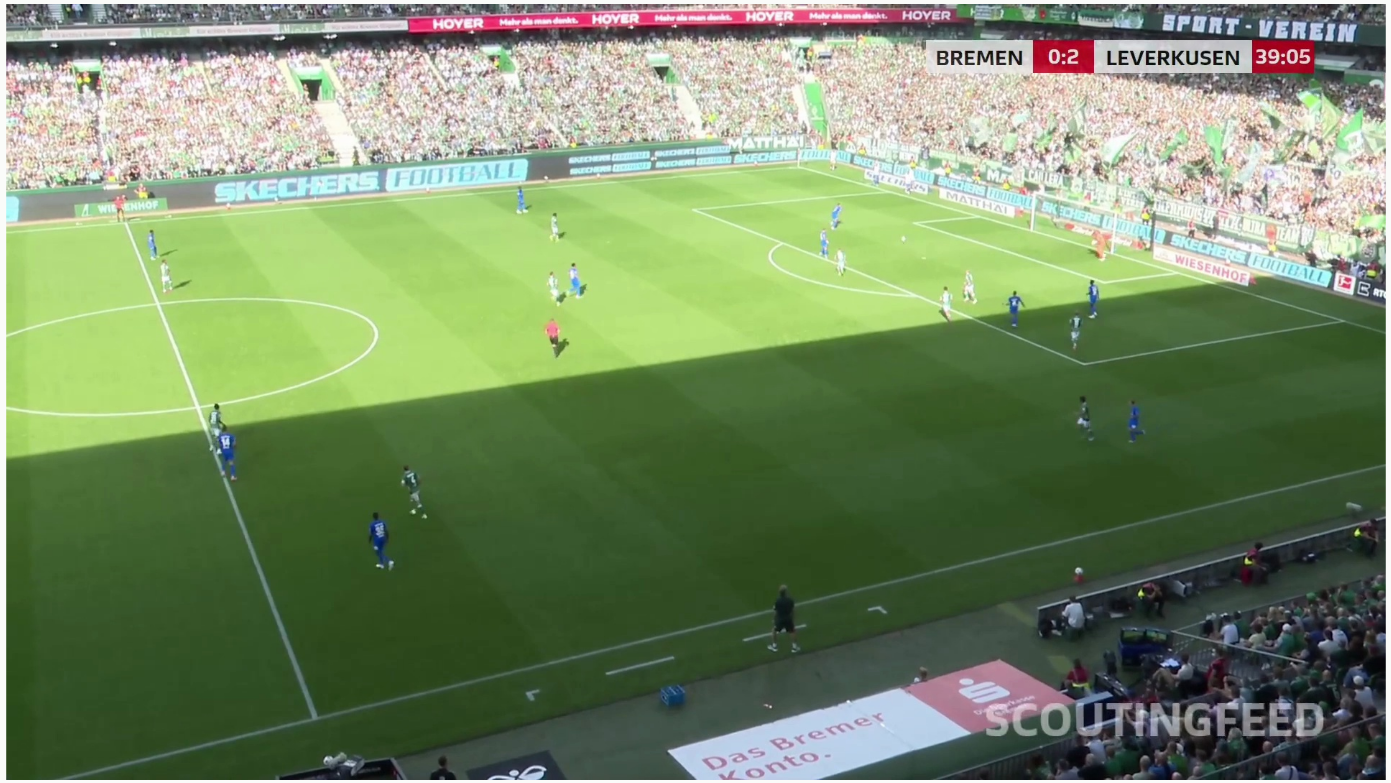
Controlled experiments in football video



Master’s thesis
Scientific design and current status

Research question

How does preprocessing change pose quality when model weights stay fixed?



Independent variable

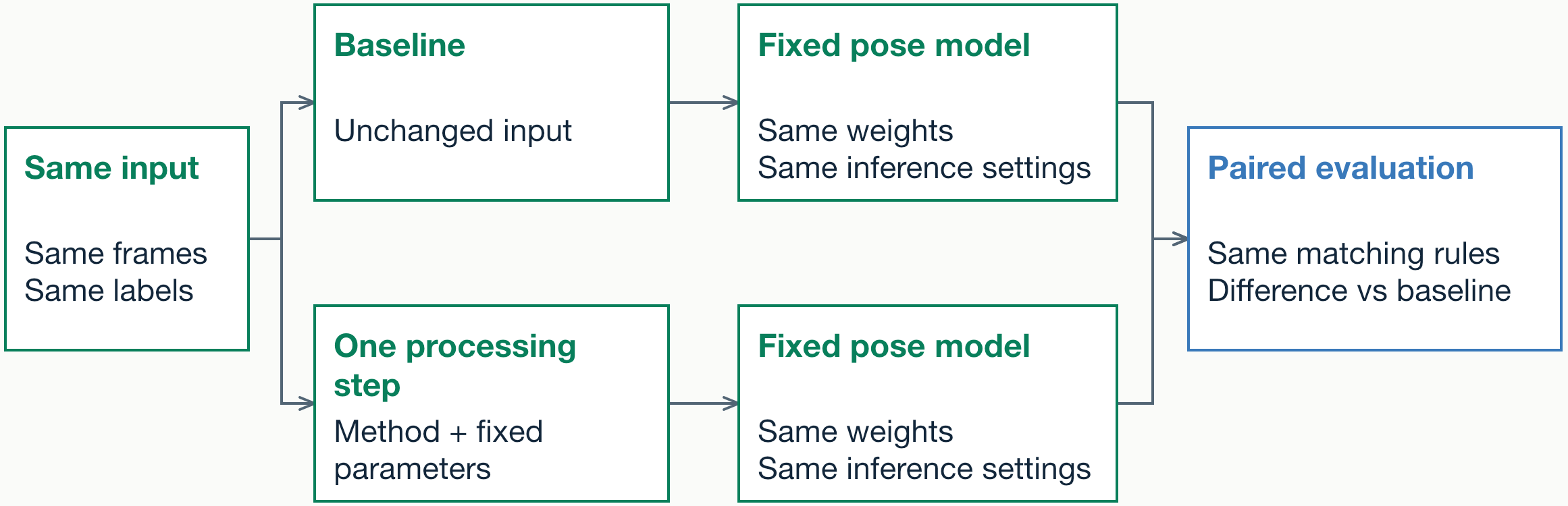
Processing method and its parameters

Measured outcomes

Detection coverage
Joint localization
Computational cost

Controlled comparison

Fixed published pretrained checkpoints. No training or fine-tuning within the comparison.



Ground truth for distinct evaluation tasks

Reference data must describe the quantity that each metric evaluates.

Evaluation task	Independent reference	Current position
COCO pose benchmark	Official person keypoints and visibility labels	Available
Football image-plane pose	Validated DFL/Werder joints, if available Otherwise reviewed manual annotations	Format and alignment still pending
Pitch-coordinate positions	Synchronized tracking x/y with independently checked calibration	Later evaluation stage

Football reference-data protocol

Proposed workflow before the final accuracy experiment

1. **Define the evaluation sample**

   Include varied scale, lighting, occlusion and camera views

2. **Align the reference to the video**

   Match timestamps, player identities and joint definitions

3. **Review labels independently**

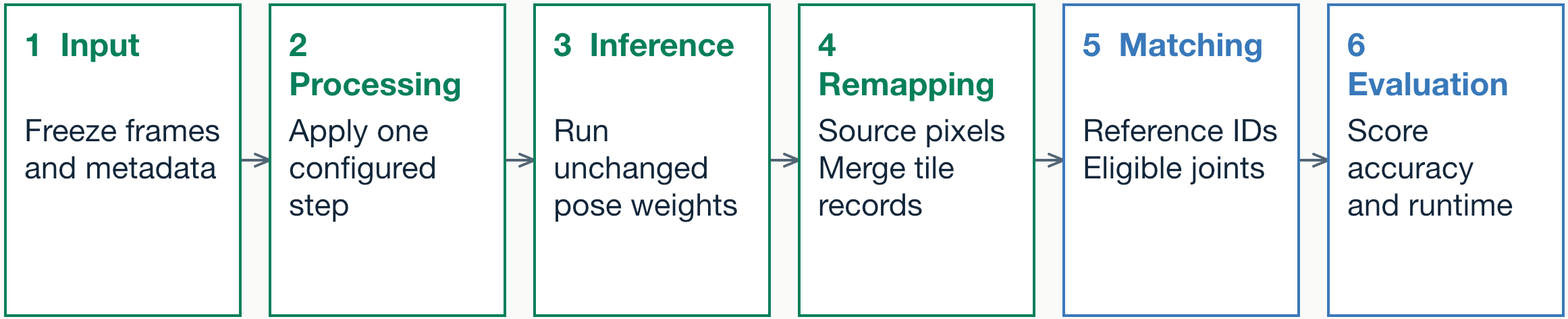
   Record visibility, uncertainty and annotation disagreements

4. **Freeze development and test sets**

   Separate video sequences and version the approved labels

Pipeline, step by step

Every prediction returns to the original frame before matching against ground truth.



Evaluation metrics

COCO AP/AR scoring is implemented. Football accuracy metrics require aligned ground truth.

Metric	Question it answers
AP / AR using OKS	How accurately and completely do we recover person poses?
PCK and PCK-AUC	How often do joints fall within a normalized distance tolerance?
PCP	Do the endpoints of a body part satisfy a fixed tolerance?
Player precision / recall	How many detections are correct, and how many targets do we find?
Detected-joint coverage	How many eligible reference joints receive confident predictions?
Pitch-position error	How far are player ground points from tracking positions, in metres?

Estimating the processing effect

Paired comparisons use the same reference population in every condition.

$$\Delta \text{ score} = \text{score(processed input)} - \text{score(baseline)}$$

1. **Fixed denominators**
   Count missed targets and unpredicted eligible reference joints
2. **Separate error sources**
   Report detection, pose localization and pitch-filter errors
3. **Independent confirmation**
   Choose settings on development clips, then test held-out sequences

Reproducibility safeguards

A result needs a recoverable input, configuration and execution environment.

Already recorded

Input and checkpoint SHA-256
Expanded YAML and processing order
Git revision and package versions
Hardware metadata and runner logs
Raw predictions and decision archives

Final-study verification

Pin actual container image digests
Freeze labels and evaluation settings
Repeat a reference run
Declare numerical tolerances
Measure cold and warm runtime separately

Experimental matrix

One unchanged baseline and eleven individual processing conditions

Processing family	Conditions
Scale and regions	Bicubic ×2, FSRCNN ×2, 2 × 2 tiling, detector-guided crops
Contrast and brightness	CLAHE, gamma darken, gamma brighten
Noise and restoration	Non-local means, bilateral denoising, unsharp, Wiener deblur

YOLO26x Pose
OpenPose BODY_25
HRNet-W32

Current football pilot
12 conditions × 3 models
782 identical source frames

Current experiment status

Exploratory detection yield on the football clip. Accuracy evaluation remains pending.

Model	Baseline on pitch	Example condition	On pitch
YOLO26	357	Bilateral denoising	2,115
OpenPose	11,647	CLAHE	15,338
HRNet	3,602	Bicubic ×2	15,554

Completed: 35 of 36 model jobs

Open cases: HRNet tiling, YOLO cropping and the failed OpenPose crop job

Next scientific milestones

1 **Reference-data agreement**

Confirm football joints, tracking timestamps and annotation fallback

2 **Protocol freeze**

Select the primary metric, matching rules and held-out sequences

3 **Verified ablation study**

Resolve runner issues, repeat baselines and score every condition

4 **Confirmation and downstream use**

Test selected combinations, then compare pitch positions with tracking

Backup: on-pitch screening overview 1/2

Record count and change relative to each model's own baseline

Processing	YOLO26	OpenPose	HRNet
Baseline	357 (reference)	11,647 (reference)	3,602 (reference)
Resize ×2	283 (-20.7%)	10,934 (-6.1%)	15,554 (+331.8%)
CLAHE	1,156 (+223.8%)	15,338 (+31.7%)	3,995 (+10.9%)
Gamma darken	315 (-11.8%)	12,193 (+4.7%)	3,634 (+0.9%)
Gamma brighten	440 (+23.2%)	11,029 (-5.3%)	3,568 (-0.9%)
NLM denoising	615 (+72.3%)	8,054 (-30.8%)	3,037 (-15.7%)

Backup: on-pitch screening overview 2/2

Record count and change relative to each model's own baseline

Processing	YOLO26	OpenPose	HRNet
Bilateral denoising	2,115 (+492.4%)	12,581 (+8.0%)	3,216 (-10.7%)
Unsharp	491 (+37.5%)	11,860 (+1.8%)	2,675 (-25.7%)
Motion deblur	1,627 (+355.7%)	7,362 (-36.8%)	214 (-94.1%)
Super-resolution ×2	312 (-12.6%)	11,583 (-0.5%)	15,565 (+332.1%)
Tiling	3,402 (+852.9%)	16,072 (+38.0%)	1 (-100.0%)
Detector crop	0 (-100.0%)	FAILED	3,467 (-3.7%)

Backup: metric conventions

Proposed football conventions must be agreed and frozen before final scoring.

Metric	Convention to document
OKS / AP / AR	Standard COCO sigmas and OKS thresholds 0.50 to 0.95 for COCO
PCK	Distance threshold and reference normalization, such as GT box diagonal
PCK-AUC	Integral over an explicitly stated PCK threshold interval
PCP	Limb list and endpoint tolerance relative to reference limb length
Coverage / precision / recall	Eligible joints, target population, confidence and one-to-one matching
Position error	Timestamp, player identity, coordinate convention and distance in metres